WORKSHEET 3.6: VECTOR & PARAMETRIC EQUATIONS OF A PLANE

2a) $\vec{m_1} = \hat{i} = (1,0,0)$
$\vec{m_2} = \hat{k} = (0,0,1)$

b) y-coordinate $= 0$.

3a) $\vec{m_1} = (-3,5,2)$ $\vec{m_2} = (-6,1,2)$

b) $\vec{m_1} = (5,-5,3)$ $\vec{m_2} = (1,6,-2)$

c) $\vec{m_1} = (4,-2,1)$ $\vec{m_2} = (-1,5,2)$

} as long as $\vec{m_1} \not\parallel \vec{m_2}$

4a) $\vec{r} = (9,4,-3) + t(-2,2,1) + p(0,-2,6)$

if $t=0, p=0$

$p + (9,4,-3)$

if $t=1, p=1$

$x = 9-2(1) = 7$
$y = 4+2(1)-2(1) = 4$
$z = -3+(1)+6(1) = 4$

} $p + (7,4,4)$

b) $\vec{r} = (0,1,0) + t(1,0,-2) + p(0,0,4)$

if $t=0, p=0$

$p + (0,1,0)$

if $t=1, p=0$.

$x = 0+(1) = 1$
$y = 1 = 1$
$z = 0-2(1) = -2$.

} $p + (1,1,-2)$

c) $x = 3+5s+t$
$y = -2-5s+6t$
$z = -2+3s-2t$

if $s=0, t=0$

$p + (3,-2,-2)$

if $s=1, t=0$

$x = 3+5 = 8$
$y = -2-5 = -7$
$z = -2+3 = 1$

} $p + (8,-7,1)$

d) the xz-plane, any as long as y-coordinate $= 0$

$pt\,(1,0,0)$ $\quad$ $pt\,(1,0,1)$

5a)
$$\left.\begin{aligned} x &= -4+5s-4t \\ y &= -6+2s-6t \\ z &= 3+3s+3t \end{aligned}\right\} \; s,t \in \mathbb{R}$$

b)
$$\left.\begin{aligned} x &= 3t \\ y &= 2s \\ z &= 1 \end{aligned}\right\} \; s,t \in \mathbb{R}$$

c) $\vec{r} = (0,0,0) + s(1,0,0) + t(0,0,1)$, $s,t \in \mathbb{R}$

$$\left.\begin{aligned} x &= s \\ y &= 0 \\ z &= t \end{aligned}\right\} \; s,t \in \mathbb{R}$$

6a) $\vec{r} = (-4,-1,3) + s(1,3,4) + t(3,-4,-1)$, $s,t \in \mathbb{R}$

b) $\vec{r} = (0,4,0) + s(7,0,0) + t(0,0,-2)$, $s,t \in \mathbb{R}$

c) $\vec{r} = (0,0,0) + s\underbrace{(1,0,0)}_{\hat{i}} + t\underbrace{(0,0,1)}_{\hat{k}}$, $s,t \in \mathbb{R}$

7a) $\vec{r} = (-4,5,1) + s(-3,-5,3) + t(2,-1,-5)$ $\quad s,t \in \mathbb{R}$

b) since $\vec{m_1} \not\parallel \vec{m_2}$

$\vec{r} = (4,7,3) + s(1,4,3) + t(-1,-1,3)$, $s,t \in \mathbb{R}$

c) $pt\,(-3,4,6)$

$\vec{m_1} = (-5,-2,3)$

$\vec{m_2} = \vec{AB} = (8,3,5) - (-3,4,6) = (11,-1,-1)$

$\therefore \vec{r} = (-3,4,6) + s(-5,-2,3) + t(11,-1,-1)$ $\quad s,t \in \mathbb{R}$

d) pt $(0, 1, 3)$

$\vec{m_1} = (4, 2, -4)$

$\vec{m_2} = (0, 1, 3) - (-4, 5, -4)$

$= (4, -4, 7)$

$\therefore \vec{r} = (0, 1, 3) + s(4, 2, -4) + t(4, -4, 7)$, $s, t \in \mathbb{R}$

e) pt $(2, 6, -5)$

$\vec{m_1} = (2, 6, -5) - (-3, 1, -4)$

$= (5, 5, -1)$

$\vec{m_2} = (2, 6, -5) - (6, -2, 2)$

$= (-4, 8, -7)$

} any scalar multiples of $\vec{AB}$, $\vec{BC}$, $\vec{AC}$, etc.

$\therefore \vec{r} = (2, 6, -5) + s(5, 5, -1) + t(-4, 8, -7)$ $s, t \in \mathbb{R}$

8a) $x = 7 + 4s - 3t$
$y = -5 - 5 + 4t$
$z = 2 + s + 4t$
} $s, t \in \mathbb{R}$

b) $x = 5 - 3s + 4t$
$y = 4 + s - 2t$
$z = 2 + 4s + t$
} $s, t \in \mathbb{R}$

c) pt $(1, 3, -1)$

$\vec{m_1} = (2, 2, -5)$

$\vec{m_2} = (8, 3, 5) - (1, 3, -1)$

$= (7, 0, 6)$

$x = 1 + 7s + 2t$
$y = 3 + 2t$
$z = -1 + 6s - 5t$
} $s, t \in \mathbb{R}$

d) pt $(3, 2, 2)$

$\vec{m_1} = (6, -4, 4)$ OR$_{//}$ $(3, -2, 2)$

$\vec{m_2} = (3, 2, 2) - (1, 6, -6)$

$= (2, -4, 8)$ OR$_{//}$ $(1, -2, 4)$

$x = 3 + 3s + t$
$y = 2 - 2s - 2t$
$z = 2 + 2s + 4t$
} $s, t \in \mathbb{R}$

e) pt $(2, 6, -5)$

$\vec{m_1} = (2, 6, -5) - (-3, 1, -4)$

$= (5, 5, -1)$

$\vec{m_2} = (2, 6, -5) - (6, -2, 2)$

$x = 2 + 5s - 4t$
$y = 6 + 5s + 8t$
$z = -5 - 5 - 7t$
} $s, t \in \mathbb{R}$

9a) $\vec{r} = (6,4,2) + s(0,1,0) + t(0,0,1)$, $s,t \in \mathbb{R}$

b) $P + (0,0,0)$

$\vec{m_1} = (3,3,3)$

$\vec{m_2} = (8,-1,-1)$

$\vec{r} = (0,0,0) + s(3,3,3) + t(8,-1,-1)$, $s,t \in \mathbb{R}$

c) any point on x-axis , eg $(0,0,0)$ origin.

$\therefore \vec{m_1} = (1,0,0) \Rightarrow \hat{i}$ rep x-axis.

$\vec{m_2} = (-1,-4,-7)$

$\therefore \vec{r} = (0,0,0) + s(1,0,0) + t(-1,-4,-7)$, $s,t \in \mathbb{R}$

10a) Let $A(2,3,-1)$

$B(8,5,-5)$

$C(-1,2,1)$

$\vec{AB} = (6,2,-4)$

$\vec{AC} = (-3,-1,2)$

} scalar multiples.

$\therefore$ points are collinear; not enough to determine a plane.

b) $\vec{r} = (4,9,-3) + t(1,-4,2)$

create a vector between $(8,-7,5)$ & $(4,9,-3)$

$\vec{a} = (4,-16,8)$

$\vec{m} = (1,-4,2)$

} scalar multiples.
$\therefore$ not enough to determine a plane

also ...

$8 = 4 + t$

$4 = t$

$-7 = 9 - 4t$

$-16 = -4t$

$4 = t$

$5 = -3 + 2t$

$8 = 2t$

$4 = t$

$\therefore$ $(8,-7,5)$ is a point on the line !!

11)

$$\left.\begin{aligned} x &= 7-t \\ y &= -2t \\ z &= -7+t \end{aligned}\right\} t \in \mathbb{R}$$

does not intersect z-axis.

$\therefore$ // to z-axis.

$\therefore$ $(0,0,1)$ as a direction vector

$$\therefore \vec{r} = (7,0,-7) + s(0,0,1) + t(-1,-2,1), \quad s,t \in \mathbb{R}$$

$$\left.\begin{aligned} x &= 7-t \\ y &= -2t \\ z &= -7+s+t \end{aligned}\right\} s,t \in \mathbb{R}$$

12)

$$\vec{r} = (a,b,c) + s(d,e,f) + t(a,b,c), \quad s,t \in \mathbb{R}$$

If passes through $(0,0,0)$, then $\exists$ s,t satisfy the equation

$$0 = a + ds + at \Rightarrow -a = ds + at \quad ①$$
$$0 = b + es + bt \Rightarrow -b = es + bt \quad ②$$
$$0 = c + fs + ct \quad ③$$

① $\times e$ $\quad -ae = eds + aet$

② $\times d$ $\quad -bd = eds + bdt$

$$-ae + bd = aet - bdt$$
$$-(ae - bd) = (ae - bd)t$$
$$\frac{-(ae-bd)}{(ae-bd)} = t$$
$$-1 = t$$

sub into ①

$$-a = ds + a(-1)$$
$$-a = ds - a$$
$$\phi = ds$$
$$\phi = s$$

check in ③

$$LS = \phi \qquad RS = c + f(\phi) + c(-1)$$
$$= c - c$$
$$= \phi$$

$\therefore LS = RS.$

$\therefore s = \phi, \ t = -1$

If $s=0, \ t=-1$

$$\vec{r} = (a,b,c) + \phi(d,e,f) - 1(a,b,c)$$
$$= (0,0,0) \ \checkmark$$